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Final Project Proposal: Optimal Resource Allocation Policies for Binding Free Energy Calculations in Drug Discovery

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1. Goal

Drug discovery is a search over an enormous space of candidate molecules, where the cost of evaluating any one of them in the lab makes exhaustive testing impossible. Advanced computational methods have reduced this work; via molecular dynamics simulations to predict how well a potential drug binds to a biological target (protein or RNA), teams can rank candidates and synthesize only the most promising options. Today, one method used to rank candidates is to estimate the free energy of binding between a drug and its target: the difference between the free energy of the bound and unbound state.

But these calculations are far from free. Binding is a rare event, occurring on microsecond timescales that would require roughly 10^9 timesteps to reach in a direct simulation. Instead, the field uses **alchemical free energy calculations** (Ghidini et al., 2025). Free energy is a state function, and thus the difference between two states is path independent. We construct a path that is physically impossible but easier to work with. A coupling parameter λ gradually "turns off" the interactions of the ligand until it becomes a non-interacting ghost, with the potential modeled by a simple interpolation:

$$V(\mathbf{r}, \lambda) = (1 - \lambda)\, V_{\text{on}}(\mathbf{r}) + \lambda\, V_{\text{off}}(\mathbf{r}) \tag{1}$$

Here $\mathbf{r}$ denotes the configuration of every atom in the system. The energy difference along our path can then be written as an integral:

$$\Delta G = \int_0^1 \left\langle \frac{\partial V}{\partial \lambda} \right\rangle_\lambda d\lambda \tag{2}$$

where the angle brackets denote an average over multiple configurations sampled from a Boltzmann distribution given a fixed value of λ. The average over configurations requires running a simulation at that λ. The integral is evaluated numerically over a discrete set of $\approx$ 20–40 values, each requiring an independent molecular dynamics simulation called a "window". In practice, each window is run as several independent trials from different initial conditions.

How long each trial runs and how many trials are launched are both choices. The simplest policy uses a uniform number of trials per window with equal simulation time. A more sophisticated option, used by some modern workflows, is a greedy policy that allocates additional simulation time to whichever windows show the largest remaining uncertainty in their estimate of $\langle \partial V/\partial \lambda \rangle$, aiming to minimize total variance (Clark et al., 2024).

It is not clear in literature whether these policies are enough on harder systems, where some windows converge slowly and their estimates are unreliable. I want to investigate a wider set of allocation policies on a sufficiently difficult ligand system. To compare policies, I will measure how many time steps were required to reach a set target error.

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2. Decision Making

The calculation proceeds in rounds. At any moment we have a partial estimate of the binding free energy, with an error attached to each window. An action is to spend the next block of simulation time on one window, either continuing one of its trials or starting a new trial for that window, or to stop and report the current estimate. The observation is the new samples that come back, which update that window's estimate and typically shrink its error. The reward is a cost for every unit of time spent, plus a large penalty for stopping while the answer is still wrong. What we cannot see is how noisy each window really is, so the errors we are working from are themselves estimates. This is not a one-shot since our choices depend on the ever-changing progress in each window.

3. Sources of Uncertainty

First, we don't know how noisy each window is, which decides how much time it needs. We only learn this by sampling, so early on our errors are rough estimates, and this error should be modeled with a probability distribution. Second, the samples themselves are random: the same window given the same time returns different numbers each run, so policies have to be compared using random seeds. Third, a window's computed error can be small while its value is wrong, if its trials have not sampled the system properly. I plan to model the smoothness of $\langle \partial V / \partial \lambda \rangle$ across λ, since neighboring windows describe nearly the same system and should give similar values, which gives a way to identify outliers.

4. Sketches of Solution

The first step is to find a system where these decisions matter. I plan to select a harder transformation of the kind used to stress-test recent methods (Lee et al., 2025) and run every window uniformly and far longer than any policy would need. That gives a converged answer to treat as ground truth. No further simulation is required, since a policy's decision to spend time on a window just becomes a decision to reveal more of that window's recorded samples. This makes it cheap to compare many policies on identical data across many random seeds, scoring each on how many timesteps it needed to reach the target error.

References

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